A Practical Hybrid Robust VAE-CatBoost Framework for Efficient Multiclass Network Intrusion Detection

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Abstract. Network intrusion models frequently achieve high scores on random data splits yet miss attack families excluded from training. This paper introduces a hybrid architecture in which a robust variational autoencoder (VAE) with a Student-t likelihood, explicit masking of undefined flow statistics, and a topological loss feeds a CatBoost classifier and two online modules. From the VAE's negative log-likelihood and latent coordinates, a streaming correction relabels bursts of novel flows with the most frequent recent attack prediction, while a secondary detector flags unseen attack families. On CSE-CIC-IDS2018 with a chronological hold-out across three seeds, our approach attains a mean per-class F1 of 0.991 ± 0.004 over the four larger classes, versus 0.979 ± 0.001 for raw-feature CatBoost and 0.975 for the strongest of ten untuned re-run methods. The same VAE-driven correction raises raw-feature CatBoost to 0.990 too, confirming that the correction on the VAE signals drives the gain. Trained on attacks from other days, the framework with the detector achieves a Youden's J of 30.27 % on left-out days, against 6.05 % for the best of five re-run methods trained on the same days, and the detector alone flags 30.86 % of an unseen family's flows, versus 4.81 % on raw features without monotone constraints. Single-thread CPU inference of the full pipeline takes 0.05 ms per flow.

Keywords: Intrusion detection · Variational autoencoder · Student-t likelihood · CatBoost · Novelty · Unseen attack families

1 Introduction

Flow-based intrusion detection systems (IDS) must handle skewed classes, extreme outliers in traffic statistics, and attack families absent from training. Hybrids of feature selection, autoencoders, and gradient-boosted trees report recalls near 99 % [1] and accuracies near 98 % [2] on public flow datasets, but such figures can be optimistic: random partitioning places flows of one attack session in both training and test sets, port numbers let a model memorize attacker hosts, and accuracy obscures weak rare classes [3–6]. Furthermore, a closed-set classifier outputs only known labels, so its detection can collapse on a new attack family [7, 8].

We retain the multi-stage hybrid design while redefining the autoencoder's function. In our framework (Fig. 1), Stage 1 selects features using CatBoost SHAP values,

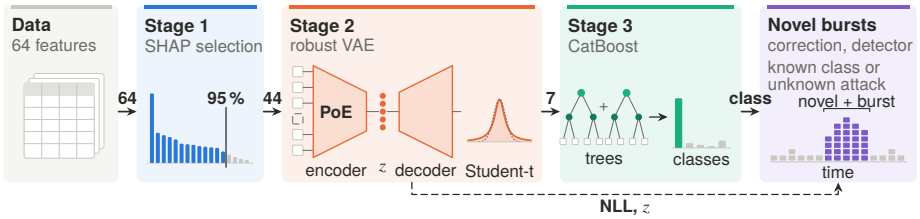


Fig. 1. Architecture: Stage 2 encodes the defined features with a product of per-feature experts (PoE; an undefined statistic, dashed, is left out). In Stage 3, the correction relabels bursts of novel flows from the VAE NLL and latent mean z, and the detector checks the remaining Benign flows.

Stage 2 encodes them through a robust variational autoencoder (VAE), and Stage 3 classifies each flow with CatBoost from the VAE outputs. The VAE's reconstruction error quantifies how unusual a flow is, while its latent space groups similar traffic; Stage 3 feeds both signals into an online burst correction and an unseen-attack detector. We call Stages 1–3 without the detector *the framework*, and write *with the detector* when it is added. Our contributions are:

1. A robust VAE design for CICFlowMeter flow features that treats undefined statistics as missing entries rather than zeros, models heavy tails with a Student-t likelihood, and adds a topological loss for latent neighbourhoods; its NLL and latent space feed Stage 3 (Sect. 3).
2. A streaming correction that relabels bursts of novel flows and a monotone CatBoost detector for unseen families; both run online without host addresses (Sect. 3).
3. An evaluation on CSE-CIC-IDS2018 on a hold-out, left-out days, leave-one-attack-out folds, and a test day against ten re-run methods, with ablations and CPU timing (Sect. 4).

2 Related Work

Detection varies widely across prior studies and can drop sharply when test attack classes are absent from training [7–10] (Table 3, bottom). Earlier hybrids feed autoencoder errors [1] or latent VAE representations [2] into tree classifiers, whereas streaming detectors keep per-host state [11], score edge bursts between host pairs [12], or classify multi-flow windows with transformers [13]. Our streaming components instead track bursts within the VAE latent space and need no IP addresses.

3 Proposed Framework

3.1 Stage 1: CatBoost with SHAP Feature Selection

Stage 1 ranks the 64 candidate features by mean absolute SHAP value [14] across 20 CatBoost runs, each on a random 70 % of the features, and retains the smallest top set holding 95 % of the importance: 44 features on our data (Fig. 2, Table 1).

3.2 Stage 2: Robust Variational Autoencoder

Each part of our VAE [15] addresses a property of flow statistics.

Undefined Statistics. When CICFlowMeter cannot compute a statistic—such as an inter-arrival time from a single packet or an idle time without an idle period—it writes a default value of 0 [16]. These zeros represent missing entries rather than measurements; in the training rows, 81.04 % of *Idle Max* values and 28.82 % of *Flow IAT Std* values are undefined. We follow the CICFlowMeter source code to flag these entries as missing ($b_j = 0$). Each feature j has its own small encoder, an expert, mapping x_j to a Gaussian $\mathcal{N}(\mu_j, \sigma_j^2)$ over the latent vector $z \in \mathbb{R}^5$. The posterior is the product of the defined-feature experts and a standard normal prior expert [17]: $\sigma^{-2} = 1 + \sum_j b_j \sigma_j^{-2}$ and $\mu = \sigma^2 \sum_j b_j \sigma_j^{-2} \mu_j$. Thus, a missing feature neither affects the posterior nor adds a likelihood term [18, 19].

Heavy Tails. Because training flow durations span six orders of magnitude and forward packet counts span four, a few extreme outliers would otherwise dominate a standard Gaussian likelihood. The decoder therefore assigns each feature a Student-t likelihood with learned location, scale, and degrees of freedom [20], with negative log-likelihood ℓ_j. A soft lower bound on the log scale prevents the scale from collapsing [21].

Neighbourhoods. Stage 3's burst correction counts flows within latent grid cells, which requires similar traffic to remain close after encoding—a property that the standard VAE objective does not enforce. We therefore add the 0-dimensional topological loss $\mathcal{L}_t$ of Moor et al. [22], comparing the minimum spanning trees of the inputs and posterior means within each batch.

Objective. To train the experts on incomplete rows, each batch also includes a second view x_s per row that retains a random subset s of the defined features [18, 19]. The VAE minimizes

$$\mathcal{L} = \sum_{v \in \{x, x_s\}} \Big(\mathbb{E}_{q(z|v)} \big[\textstyle\sum_j m_j^v \ell_j \big] + \beta \, \mathrm{KL}\big(q(z \mid v) \,\|\, p(z)\big) \Big) + 0.1\beta \, \mathrm{KL}\big(q(z \mid x_s) \,\|\, q(z \mid x)\big) + \gamma \mathcal{L}_t, \quad (1)$$

where m_j^v is 1 for features present in view v, $p(z) = \mathcal{N}(0, I)$, and the third term aligns the two views [23]. The weight is $\gamma = 1$ (Sect. 3.4).

3.3 Stage 3: Classification, Correction and Detection

Classifier. For each flow, the VAE produces seven outputs: the five posterior means μ, the KL term, and the reconstruction NLL at $z = \mu$, summed over the 23 features defined for every flow so that the sum has the same terms across flows. A multiclass CatBoost classifier [24] with balanced class weights predicts the class from these outputs and is tuned with Optuna [25]. Since the classifier outputs only training labels, evaluations on unseen attack families treat any non-Benign prediction as a detected attack, following the binary convention of [7].

Novel-Burst Correction. A learned attack can recur in a form the classifier misses: Slowloris teardown flows appear Benign to the classifier, yet look unusual to the VAE and arrive in bursts. The correction identifies such flows in three steps. First, a flow is *novel* if its reconstruction NLL exceeds the q_n quantile of Benign NLLs on a calibration day. Second, a flow is part of a *burst* if many flows have recently arrived in its latent cell: after dividing each latent axis by its training SD, the cell is the grid cube of side w that contains the flow's posterior mean. Because the flow records lack IP addresses, MIDAS-R [12] evaluates the latent cell in place of a host pair, flagging a burst whenever its score exceeds the q_b quantile of Benign training flows. Third, a novel flow inside a burst adopts the attack class most frequently predicted by the classifier among earlier flows over the preceding W seconds of the same day, provided that this class count clears a support threshold; otherwise, its prediction remains unchanged. Since the calibration day contains no training attack classes, any attack prediction on its Benign flows is a false alarm, and the q_s quantile of these false-alarm counts sets the threshold. Every step uses only earlier flows, so the correction runs online. During the hold-out evaluation, same-day training flows remain in the stream and count as Benign predictions, so that no training label adds attack support.

Unseen-Attack Detector. A flow from an unseen attack family that triggers no known-class prediction remains Benign after the burst correction, so a secondary CatBoost model, the detector, examines every remaining Benign flow. Because the normal NLL level varies across capture days, its first input is the *relative NLL*: the reconstruction NLL minus the running median NLL of earlier non-burst flows from the same day. The four additional inputs are the KL term, the latent cell's MIDAS-R score, a novel-only MIDAS-R score over flows whose relative NLL exceeds its q_n quantile on calibration-day Benign flows, and the fraction of novel flows in the cell over the last W seconds. Converting each input to its percentile rank among Benign reference flows puts all days on a shared scale. Monotone constraints [26] ensure that a larger input never lowers the attack score and make it harder for the tree model to learn a narrow value range of one training family.

Its positives are labelled attacks from families absent from the VAE and classifier training labels: the training rows of the left-out class in each leave-one-attack-out fold, encoded by that fold's models, plus the attacks of day 02-28 and further capture days (Sect. 4.1). The negatives are half of the Benign flows from these days and the calibration day, with the cut-off set for 1 % false alarms on the other half. A left-out day excludes that day and its families on all days from detector training; a fold excludes its class. Any flagged flow becomes an *unknown attack*, a wrong class in the seen-class metrics.

3.4 Hyperparameter Selection

No model weight or threshold is fitted on the hold-out or test day, and the VAE stops on the 02-28 ELBO. The topological weight γ and the five correction settings are chosen with seed 3 on an inner split holding out the latest 20 % of training rows per day and class. The weight γ maximizes mean per-class F1 with the correction at its start values. The correction settings are then tuned one by one to maximize mean attack-versus-Benign F1 across three leave-one-attack-out folds of the inner split. For the detector, the NLL

Table 1. Settings of the framework; Stage 3 values are search ranges of Optuna's TPE sampler.

Part	Settings
Stage 1	20 CatBoost runs on random 70 % feature subsets and ≤50,000 training rows per class, SHAP on 1,000 rows per class; 300 iterations, depth 6, learning rate 0.1, balanced class weights
Stage 2	all training rows; signed log, standardization on defined entries; 44 experts of 20 ReLU units, latent size 5; decoder 128 ReLU units, Student-t, soft bound $\log \hat{\sigma} \geq -6$, $\nu = 1 + \text{softplus}$; $\beta = 0.5$, consistency 0.1β; second view: each defined feature kept with probability $1 - \rho$, $\rho \sim U(0, 0.7)$, and, in addition, the features removed that are undefined in a random other row of the batch; $\gamma \in \{0.1, 0.3, 1, 3\}$, chosen $\gamma = 1$, on 256 flows per batch, distances divided by their mean; Adam $8 \cdot 10^{-3}$, batch 32,768, ≤1,000 epochs, learning rate halved after 3 flat epochs, stop after 10 epochs without a relative gain of 10^{-4} in the 02-28 ELBO
Stage 3	CatBoost, plain boosting, balanced weights; Optuna TPE, 20 trials (10 random); iterations 200–1000, depth 4–8, learning rate 0.01–0.3, L2 1–30 (both log scale); mean per-class F1 over 3 time-blocked folds of a 10 % sample per day and class (classes under 2,000 rows whole); same search for raw CatBoost
Correction	grids, searched in this order from the start values in brackets: $w \in \{0.1, [0.2], 0.5\}$, $q_b \in \{0.975, [0.99], 0.995\}$, $W \in \{30, [60], 120\}$ s, $q_s \in \{0.975, [0.99], 0.995\}$, $q_n \in \{0.90, 0.95, [0.975], 0.99\}$; chosen $w = 0.1$, $q_b = 0.975$, $W = 120$ s, $q_s = 0.975$, $q_n = 0.9$; latent axes divided by their training SD; MIDAS-R edge score only, decay 0.5 [12], one-minute ticks; flows of the same second are not earlier than each other
Detector	CatBoost, depth 2, 500 iterations, learning rate 0.1, monotone; rank references: relative NLL and KL of calibration-day Benign flows, stream inputs of Benign training flows; median NLL of a reservoir sample of ≤20,000 earlier flows of the day with MIDAS-R below its 0.9 Benign training quantile, refreshed every 500 flows (calibration-day median for the first 1,000); ≤200,000 negatives per day; 1 % false alarms

is taken relative to either the calibration-day median NLL or the day's running median, and the inputs are chosen from five subsets; each configuration is validated by leaving out one further capture day or one fold at a time. The chosen configuration—the running median with all five inputs—achieves the highest mean detection rate among those with at most 2 % mean false alarms. Table 1 lists the grids and selected values.

4 Experiments and Results

4.1 Dataset and Protocol

We used the official CSE-CIC-IDS2018 CSV files [27, 28], corrected their 12-hour CICFlowMeter timestamps [16], and removed rows with non-finite values. To prevent shortcut learning, we removed the timestamp [4, 6], the destination port [5], 10 constant columns, and three TCP header fields that identify the attacker hosts on the training days (*Init Fwd Win Byts*, *Init Bwd Win Byts*, *Fwd Seg Size Min*; similar shortcuts are discussed in [4, 6]), leaving 64 candidate features. Exact duplicates, cross-day matches of training rows, and label conflicts (identical feature vectors with different labels) were filtered out; the conflicts include 184,551 FTP brute-force rows, consistent with the failed FTP attack in [29].

Table 2 summarizes the resulting splits. Test day 02-21 contains only attack families excluded from training and serves as our unseen-family test set. Although its HOIC window contains a single-template burst of 358,621 Benign flows, Liu et al. [29] noted no HOIC labelling problem beyond a time offset, so we retained these flows as Benign; only 2,125 Benign flows of that day occur outside this burst. The detector also trains on further capture days in which at least six distinct hours each hold at least 1 % of the day's Benign flows, a rule that excludes the truncated file of day 02-16. On day 02-20,

Table 2. Data splits after cleaning: the hold-out contains the latest 20 % of each day and class.

Split (use)	Day(s)	Rows (class counts)
Training (Stages 1–3)	02-14, 02-15	1,039,918 (Benign 923,868; SSH 75,224; GoldenEye 33,066; Slowloris 7,730; FTP 30)
Hold-out (seen classes)	02-14, 02-15	259,983 (Benign 230,968; SSH 18,807; GoldenEye 8,267; Slowloris 1,933; FTP 8)
Validation (VAE stop, detector)	02-28	222,123 (Benign 211,538; Infiltration 10,585)
Calibration	02-23	705,316 (Benign 704,897; web attacks 419)
Test (unseen classes)	02-21	555,189 (Benign 360,746; HOIC 192,713; LOIC-UDP 1,730)
Further days (detector)	02-20, 02-22, 03-01, 03-02	6,659,249 (Benign 5,896,584; LOIC-HTTP 573,708; Bot 143,315; Infiltration 45,408; web brute force 147; XSS 64; SQL injection 23)

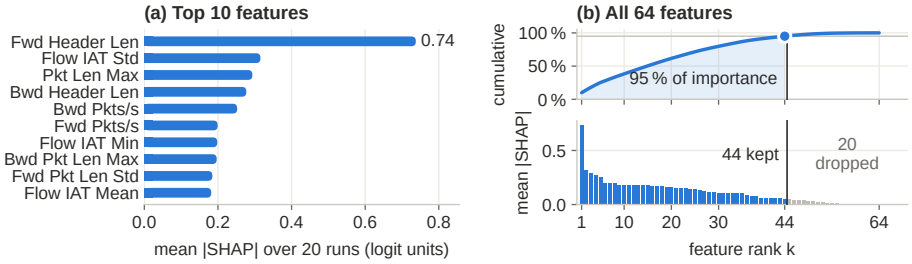


Fig. 2. Stage 1 SHAP ranking: (a) mean |SHAP| of the top 10 features over 20 runs; (b) cumulative share and importance of all 64 features (the 95 % rule keeps $k = 44$). Dividing the 20 runs into two disjoint halves yields a Spearman rank correlation of 0.963 and a top-44 Jaccard index of 0.913.

we removed 797 UDP flows labelled LOIC-HTTP within the LOIC-UDP window, since LOIC-UDP is a test-day class.

Metrics and Runs. Seen classes are evaluated by the mean per-class F1 over the classes with at least 30 hold-out rows, with the 8-row FTP class reported separately. For unseen families, we track the detection rate (DR), the Benign false-positive rate (FPR), and Youden's $J = \mathrm{DR} - \mathrm{FPR}$. Stage 1 feature selection runs once on the main split and once per fold, whereas all later models and thresholds are retrained for each of the seeds 3, 4, and 5. We report the mean and standard deviation (SD) over seeds alongside a paired bootstrap of the difference to raw CatBoost [30]. The VAEs and PyTorch re-run methods were trained on an RTX 5080 Laptop GPU and the other models on an Intel Core Ultra 9 275HX CPU; the code repository gives library versions, thread counts, and the command for every table: https://github.com/WinterBlossom0/IDS-VAE-and-Catboost.

Comparison Methods. Ten methods from related work—the *re-run methods*—were re-run with the same Stage 1 features of the main split or fold, the same rows, and the same seeds (Table 3), using class-balanced weights or class-balanced batches without synthetic oversampling. Each is evaluated on the hold-out, the test day, and the further capture days, and is retrained without the left-out class in each fold.

Table 3. Re-run methods (top; deviations from the source in italics; not given: absent from the source) and reported results of related work (bottom; other splits and metrics). LOAO: leave-one-attack-out

Method	Settings
Random forest [8]	scikit-learn defaults, as the source; standardization; *signed log, balanced class weights*
MLP [7]	24 hidden units (WEKA rule of the source code); *scikit-learn with early stopping, not WEKA*
XGBoost [31], LightGBM [32]	library defaults, signed log + standardization (XGBoost has no random step)
IDAE-XG [1]	as the source; top 22 features by forest rank, 20 epochs (not given); *multiclass; no upsampling*
CNN-1D [33]	as the source ($\leq$20 epochs, patience 4); *our split, stratified random split in the source*
FlowTransformer [13]	8-flow windows (source code), 2 decoder blocks, learning rate of the source grid; *multiclass*
TabM [34]	package defaults, settings of the source (patience 16); *no tuning*
FT-Transformer [35]	default configuration and optimizer (patience 16); *no tuning, no ensemble*
TabNet [36]	Table 4 of the source, patience 20; learning rate 0.02 (library default; not given); *no SMOTE*
Work	**Data; split; reported result**
Ferreira [7]	CSE-CIC-IDS2018, train 02-16, test 02-21; MLP TPR 0.898 on test attacks, 0 on LOIC-UDP
Sakr [8]	CSE-CIC-IDS2018, 9 days; random forest F1 0.957 (random split), 0.239 (LOAO mean)
Verkerken [9]	CIC-IDS2017 random split: 41/47 unknown flows found; CSE-CIC-IDS2018 Infiltration recall 78.38 %
Soltani [10]	CSE-CIC-IDS2018 packet bytes, one unknown label per run: its accuracy 35.47–72.16 %

4.2 Seen Classes: The Correction on VAE Signals Gives the Gain

We first compare the framework with a CatBoost model trained directly on the same 44 Stage 1 features (raw CatBoost) on the four seen classes with at least 30 hold-out rows (Table 4). The framework achieves a mean per-class F1 of 0.991 ± 0.004, outperforming raw CatBoost (0.979 ± 0.001) and the strongest re-run method (0.975). Across seeds 3, 4, and 5, the margin over raw CatBoost ranges from 0.008 to 0.017, with 95 % paired bootstrap intervals of [0.006, 0.010], [0.015, 0.020], [0.009, 0.013]. Activating the detector reduces the F1 slightly to 0.990 ± 0.004, because about 1 % of the Benign hold-out flows are flagged as unknown attacks. As Table 5 shows, the gain originates from Slowloris, whereas raw CatBoost scores higher on the 8-row FTP class and therefore over all five classes (0.979 versus 0.940).

The ablation results in Table 4 pinpoint the source of this gain. Without the correction, the VAE-fed classifier trails raw CatBoost (0.965 versus 0.979). The correction raises its Slowloris recall from 0.79 (not tabulated) to 0.98 and lifts raw CatBoost to 0.990 as well, matching the framework within the seed spread. Nevertheless, raw CatBoost with the correction still runs the VAE to compute the NLL and latent cells, relabels more Benign flows, and is slower than the framework. Consequently, the gain stems from the correction using the VAE signals rather than from the VAE acting as a feature compressor. Accordingly, a Gaussian likelihood, no topological loss, or the latent means alone shift the F1 by at most 0.003 with the correction, within the seed spread, whereas without the correction the topological loss lowers the F1. On unseen families, a detector on a VAE without the topological loss reaches a left-out-day J of 26.17 %, against 26.37 % with it, when both train on the further days only, so the loss does not alter detection either. LightGBM or XGBoost in Stage 3 give a similar F1 but a lower balanced accuracy. Concatenating the 44 raw features with the seven VAE outputs still trails raw CatBoost before the correction (0.973 versus 0.979), ties the framework after it (0.991), and removes the FTP false alarms (FTP F1 0.98 versus 0.74).

Table 4. Hold-out mean per-class F1 (classes ≥30 rows) per seed and mean ± SD, balanced accuracy (all classes), Benign hold-out flows relabelled (sum over seeds; with the detector: by the correction and the detector), and CPU ms per flow (batch 1, one thread, seed 3). With the correction, the framework has a higher mean F1 than raw CatBoost and each re-run method.

Configuration	Seed 3	Seed 4	Seed 5	Mean F1	Bal. acc.	Relab.	ms/flow
Ours (with the correction)	0.9883	0.9949	0.9895	0.9909 ± 0.0036	0.9947 ± 0.0018	487	0.044
Without the correction	0.9583	0.9700	0.9667	0.9650 ± 0.0060	0.9569 ± 0.0051	–	–
Without the topological loss	0.9880	0.9931	0.9906	0.9905 ± 0.0026	0.9865 ± 0.0153	405	–
No topological loss or corr.	0.9742	0.9655	0.9772	0.9723 ± 0.0060	0.9590 ± 0.0255	–	–
Gaussian likelihood	0.9876	0.9908	0.9864	0.9883 ± 0.0023	0.9940 ± 0.0008	583	–
Latent mean only (5 inputs)	0.9883	0.9913	0.9875	0.9890 ± 0.0020	0.9946 ± 0.0013	614	–
LightGBM in Stage 3	0.9905	0.9935	0.9900	0.9913 ± 0.0019	0.9767 ± 0.0284	671	0.16
XGBoost in Stage 3	0.9910	0.9954	0.9905	0.9923 ± 0.0027	0.9441 ± 0.0015	614	0.058
Raw + VAE inputs (51) + corr.	0.9902	0.9893	0.9934	0.9910 ± 0.0022	0.9950 ± 0.0007	1209	–
Raw + VAE inputs (51)	0.9781	0.9634	0.9788	0.9734 ± 0.0087	0.9673 ± 0.0085	–	–
Raw CatBoost + correction	0.9911	0.9896	0.9907	0.9904 ± 0.0008	0.9955 ± 0.0003	1377	0.057
Raw CatBoost (44 features)	0.9800	0.9776	0.9785	0.9787 ± 0.0012	0.9739 ± 0.0014	–	0.018
Ours + detector	0.9868	0.9937	0.9883	0.9896 ± 0.0036	–	7674	0.050
Re-run methods (Table 3)							
Random forest [8]	0.9642	0.9643	0.9643	0.9643 ± 0.00004	0.9042 ± 0.00004	–	0.0040
MLP [7]	0.9781	0.9555	0.9600	0.9645 ± 0.0120	0.9628 ± 0.0148	–	0.0031
XGBoost [31]	0.9640	0.9640	0.9640	0.9640 ± 0	0.9042 ± 0	–	0.0053
LightGBM [32]	0.9655	0.9677	0.9629	0.9654 ± 0.0024	0.9549 ± 0.0002	–	0.0059
IDAE-XG [1]	0.6959	0.6423	0.6425	0.6602 ± 0.0309	0.6621 ± 0.0215	–	0.016
CNN-1D [33]	0.9742	0.9780	0.9734	0.9752 ± 0.0024	0.9740 ± 0.0032	–	0.039
FlowTransformer [13]	0.8276	0.8279	0.8402	0.8319 ± 0.0072	0.8827 ± 0.0189	–	0.058
TabM [34]	0.9493	0.9510	0.9502	0.9502 ± 0.0008	0.9542 ± 0.0001	–	0.39
FT-Transformer [35]	0.9608	0.9606	0.9626	0.9613 ± 0.0011	0.9546 ± 0.0014	–	0.74
TabNet [36]	0.9546	0.9531	0.9632	0.9569 ± 0.0054	0.9594 ± 0.0058	–	0.098

Inference Time. On a single CPU thread, end-to-end processing with the detector takes 0.050 ms per flow, compared to 0.018 ms for raw CatBoost: the fused VAE and CatBoost graph accounts for 0.044 ms (0.046 ms as two separate graphs), the correction takes 0.00015 ms, and the detector takes 0.0058 ms. We timed each pipeline with a time in Table 4 as ONNX Runtime 1.31 graphs and C++ code on one thread of an Intel Core Ultra 9 275HX, one flow per call from the extracted features to the final label, over the 100,000 consecutive flows in the middle of day 02-21 (98,562 Benign burst and 1,438 attack flows). Each time is the median of 4 passes; a full repeat gave the same order of the 16 pipelines and a framework time 1.3 % higher.

Table 6 measures F1 and cost with the k most important features of Stage 1 (seed 3; $k = 64$: all 79 columns minus the 15 dropped); for $k > 10$, our F1 stays within 0.9870–0.9908, and $k \in \{23, 40, 64\}$ score above the $k = 44$ of the 95 % rule.

4.3 Unseen Attack Families

Table 7 tests the detector alone on left-out days and the framework with the detector in the leave-one-attack-out folds. The detector mainly catches families whose flows arrive as bursts of novel flows. For LOIC-HTTP, it flags 99.01 % of request flows but only 0.28 % of unanswered attempts, while Infiltration stays mostly undetected. The VAE-based inputs are key: a CatBoost detector on the 44 raw features instead of its five inputs, trained on the same rows with the same cut-off but without monotone constraints, flags

Table 5. Per-class hold-out results, mean over seeds 3–5, of the framework and raw CatBoost (top), and the confusion matrix of the framework, seed 3 (bottom; rows: true class)

		Ours (with correction)			Raw CatBoost (44 features)		
Class	Rows	Precision	Recall	F1	Precision	Recall	F1
Benign	230,968	0.9998	0.9978	0.9988	0.9989	0.9981	0.9985
DoS GoldenEye	8,267	0.9964	0.9992	0.9978	0.9983	0.9998	0.9990
DoS Slowloris	1,933	0.9801	0.9767	0.9784	0.9929	0.8724	0.9288
FTP-BruteForce	8	0.6218	1.0000	0.7360	0.9630	1.0000	0.9804
SSH-BruteForce	18,807	0.9775	1.0000	0.9886	0.9777	0.9995	0.9884
Balanced accuracy		0.9947 ± 0.0018			0.9739 ± 0.0014		
Macro ROC-AUC (one vs rest, classifier)		0.9993 ± 0.0005			0.9995 ± 0.0001		
Macro PR-AUC (one vs rest, classifier)		0.9689 ± 0.0219			0.9924 ± 0.0011		

True \ predicted	Benign	GoldenEye	Slowloris	FTP	SSH
Benign	230,374	34	56	0	504
DoS GoldenEye	8	8,259	0	0	0
DoS Slowloris	59	0	1,874	0	0
FTP-BruteForce	0	0	0	8	0
SSH-BruteForce	1	0	0	0	18,806

Table 6. Feature-count ablation (seed 3): hold-out mean per-class F1, VAE training time in s (one VAE at a time on the GPU) and inference time in ms per flow (batch 1, one thread, no detector, as Table 4)

	F1				ms per flow	
k	Ours	Raw	Raw + corr.	VAE (s)	Ours	Raw
10	0.9738	0.9815	0.9881	141	0.046	0.017
15	0.9877	0.9792	0.9886	158	0.072	0.019
23	0.9908	0.9782	0.9895	273	0.051	0.015
30	0.9870	0.9799	0.9908	357	0.054	0.024
40	0.9902	0.9813	0.9911	302	0.061	0.019
44	0.9883	0.9800	0.9911	317	0.044	0.018
64	0.9908	0.9805	0.9906	392	0.057	0.016

on average only 4.17 % (depth 6) and 4.81 % (depth 2) of each left-out family's flows, versus 30.86 % for ours (mean of Table 7a), showing that the labelled further days and tree depth alone do not explain the detection.

Table 8 compares all methods by Youden's J. Trained on days 02-14 and 02-15 only, our classifier scores like most re-run methods (J 0.57 % on left-out days); with the detector, also trained on the further days, the framework reaches 30.27 % and 78.29 % in the folds. Since the detector configuration was selected on these days and folds (Sect. 3.4), only the test day is unused by that choice. The test day shows the limit of the burst signal: 99.4 % of its Benign flows belong to one burst inside the HOIC window. The framework detects 81.35 % of LOIC-UDP flows, but also flags 66.77 % of this Benign burst, which receives higher detector scores than the HOIC flows. In the rows marked $^{+}$ in Table 8, the shallow re-run methods also train on the further days except the evaluated one, with each attack flow labelled by family, and flag flows by the detector's cut-off rule; Isolation Forest [37], an open-set detector, trains on their Benign flows only. Given the same days, the best reach J 6.05 % on left-out days (LightGBM) and 34.35 % in the folds (MLP), versus 30.27 % and 78.29 %.

Table 7. Attack families absent from all training labels, in %, mean ± SD over seeds 3–5: (a) left-out days, DR and Benign FPR of the detector alone; (b) leave-one-attack-out folds, DR on days 02-14 and 02-15 after each step, and hold-out Benign FPR; with raw CatBoost as classifier (same correction and detector), the mean final DR is 0.01 to 1.67 points higher per fold.

(a) Left-out days

Day	Family	DR	FPR
02-28	Infiltration	2.34±1.83	2.42±0.23
02-20	LOIC-HTTP	50.06±0.12	0.91±0.16
02-22	Web brute force	30.84±21.21	0.61±0.06
	SQL injection	13.04±7.53	
	XSS	48.44±17.61	
03-01	Infiltration	3.03±0.71	0.68±0.01
03-02	Bot	68.24±2.06	0.68±0.04

(b) Leave-one-attack-out folds

Removed class	Classifier	+ corr.	+ det.	FPR
SSH-BruteForce	0.0007±0.001	0.0007±0.001	95.16±5.02	2.19±1.21
DoS GoldenEye	0.11±0.01	11.96±1.40	58.19±19.27	1.68±0.92
DoS Slowloris	53.01±13.28	71.09±2.01	87.43±8.32	2.03±0.97

Table 8. Unseen attack families, in %, mean over seeds 3–5: Youden's J is the detection rate (DR) minus the Benign false-positive rate (FPR); a re-run method flags a flow whose argmax class is not Benign. $^{+}$: trained also on the further days and flagged at the detector's cut-off. Left-out days: mean over day and family; folds: mean over the three folds. Test day 02-21: DR, Benign FPR outside and inside the burst, and J against all Benign flows.

	J		Test day					
	Left-out days	Folds	DR UDP	DR HOIC	FPR out	FPR burst	J UDP	J HOIC
Ours + detector	30.27	78.29	81.35	0.01	15.42	66.77	14.88	−66.46
Random forest [8]	0.03	15.81	0	0	0.27	0.0003	−0.002	−0.002
MLP [7]	1.60	18.99	0	0	0.72	4.05	−4.03	−4.03
XGBoost [31]	−0.01	23.90	0	0	0.38	0	−0.002	−0.002
LightGBM [32]	−0.01	5.36	0.15	0	0.35	0.00009	0.15	−0.002
IDAE-XG [1]	24.57	36.52	99.81	32.17	68.17	67.64	32.16	−35.48
CNN-1D [33]	0.60	16.56	0.02	0	0.49	20.57	−20.43	−20.45
FlowTransformer [13]	8.33	8.02	93.29	17.32	38.35	68.57	24.90	−51.07
TabM [34]	0.02	16.84	63.91	0	1.36	0.57	63.34	−0.57
FT-Transformer [35]	0.31	20.85	6.61	0	0.64	0.94	5.67	−0.94
TabNet [36]	0.07	16.94	0	0.23	0.53	25.01	−24.87	−24.64
Random forest^{+}	0.16	−0.01	0	0.43	0.05	0	−0.0003	0.43
MLP^{+}	1.18	34.35	99.83	0.0002	1.38	27.43	72.55	−27.27
XGBoost^{+}	0.98	24.08	0	0.11	0.45	0	−0.003	0.11
LightGBM^{+}	6.05	22.96	0.50	0.33	0.28	0	0.50	0.33
IDAE-XG^{+}	−0.25	0.68	46.22	0.34	1.22	22.01	24.34	−21.54
Isolation Forest^{+} [37]	0.36	−0.65	66.69	0	33.44	0.01	66.48	−0.21

5 Limits

Unusual Benign bursts can trigger false alarms, attacks without bursts give no signal, and the correction assumes one dominant attack class per time window. The evidence comes from three seeds on one dataset with known errors and label faults [4, 29]. The deep re-run methods (CNN-1D, FlowTransformer, TabM, FT-Transformer, TabNet) were not given the further days, Kitsune [11] was not run, re-run methods use defaults without the correction, and missing-entry handling was not ablated.

6 Conclusion

We proposed a hybrid VAE-CatBoost framework in which a robust VAE serves both a classifier and two streaming components. On CSE-CIC-IDS2018, the correction lifts the mean per-class F1 of the four larger classes above raw CatBoost and ten untuned re-run methods, and the framework with the detector reaches a Youden's J of 30.27 % on left-out days, versus 6.05 % for the best untuned re-run method given the same days. A large Benign burst on the test day defeats the detector; separating such bursts from attacks remains future work.

Use of AI Tools. Claude Code was used for the implementation of the code and for finding bugs.

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